The Twist, version 5.1: errata and post-publication results

Danny Lee Eldridge (independent researcher) · 14 November 2026 · errata to “The Twist: a no-scale supergravity field for inflation and a thawing dark-energy tail” (v5, DOI 10.5281/zenodo.24210532)

This note lists corrections and new results found after v5 was published on 13 November 2026. The v5 paper and exploratory note are unchanged in this version; read them together with this note. A full v6 will fold these items in. Notes labelled D… are in theory/pending_docs/D72_D75_weak_spots.md of the archived repository (branch unify-bg); locked rules are in field_tests/rules/. AI-assisted.

Summary of what changed

1. **The fifth-force shield fails at one loop for the thawing tail (items 16, 12–14).** A rolling field with $K \supset -3 \ln(X + \bar{X})$ moves the Standard Model gauge couplings by $d(1/\alpha_a)/d \ln x = -b_a/4\pi$ (Kaplunovsky–Louis; fixed by rescaling anomalies), so $d \ln \alpha_{em}/d \ln x \approx 7\times10^{-3}$, far above atomic-clock bounds. Seven escape routes were tested, two with locked inflation tests (a Green–Schwarz dilaton spoils inflation; a twisted-moduli version survives only for couplings no known model provides).
2. **Adopted reading: TWIST-I, the split model (item 18).** The Twist is the inflaton; its field rolls through matter genesis and is then captured before nucleosynthesis; dark energy is a cosmological constant. This removes the shield problem and the H_0 cost. Capture without diluting the matter asymmetry needs the field to stop within $\sim10^{-7}$–10^{-5} M_P of a stabilising minimum (open model building). The original tail is kept only as a conditional variant. Locked rule: PREREG_TWISTI (+ deviations).
3. **TWIST-I’s CMB predictions are Starobinsky-class (item 19).** n_s 0.9642–0.9654 and r 0.0035–0.0038 at $N^* = 55.7$–57.6; the distinctive content is N^* from the Twist’s reheating (kination), the matter mechanism and its isocurvature, and the dark-matter window.
4. **Locked tests since v5:** the matter checks pass (yield 1.25× observed; isocurvature sensitivity $3.70 \leq 5$); the isocurvature sensitivity under the alternative coupling reading also passes ($3.68 \leq 5$, margin 1.4×).
5. **Updated costs:** full-fit $H_0 = 66.8$ km/s/Mpc for the tail (ΛCDM 68.0 on the same data); the n_s tension depends on the dark-energy background; the dark-matter candidate survives decaying-DM limits up to ~200 TeV for an O(1) gravity coupling; CMB-S4 is cancelled, and the r test now rests on the expanded Simons Observatory ($\sigma_r \approx 1.2\times10^{-3}$) and LiteBIRD ($\delta r = 0.001$, launch JFY 2032).
6. **Still running at the time of this note:** a free-neutrino-mass fit (locked, PREREG_D355_MNU_FREE) and two consistency computations (D360 T1, D367). Their results will be added to the repository.

Itemised corrections (as queued for v6)

1. Section 5.2 (Gravitino) and assumption 4: replace the text (D267, D269)

The current v5 text says: “Supersymmetry breaking comes from a separate, unspecified sector … $m_{3/2} = 10^{11}$–10^{14} GeV”.

Replace it with:

> $W = \sqrt{3}\, m\, S\, T$ vanishes at $S = 0$, so the inflaton sector does not fix $m_{3/2}$. In this field space, any supersymmetry-breaking superpotential gives $m_{3/2} = |W|/(2x)^{3/2}$ on real T [Zwirner, arXiv:2504.06190; Ellis et al., arXiv:1809.10114].
>
> **A constant W_0 is excluded.** It changes $m_{3/2}$ by a factor 5.7×10^{8} between the CMB point and today. Either the gravitino is super-Planckian during inflation, or it is $\lesssim 3\times10^{4}$ GeV today. In the second case, thermal production at $T_{RH} \approx 5.6\times10^{14}$ GeV violates nucleosynthesis bounds.
>
> Two options remain:
>
> **(i) A light gravitino from low-scale gauge mediation.** $m_{3/2} \approx 0.8$–4.7 eV today, between the Higgs-mass lower limit [Ibe & Yanagida, arXiv:1608.01610] and the cosmological upper limit [Osato et al., arXiv:1601.07386]. Along the Twist’s history it is 0.5–3 GeV during inflation ($\ll H$) and a few keV at the cliff, so it is harmless. Keeping $V \approx 0$ at every x makes the SUSY-breaking scale slide as $x^{-3/2}$ too. Superpartner masses would then change by ~30% between nucleosynthesis and today, shifting α_{em} by $\sim10^{-3}$ through thresholds (estimate). So this option needs the shield to cover the SUSY-breaking scale seen by the Standard Model (D272).
>
> **(ii) $W \supset w\, T^{3/2}$ plus a constant uplift.** This gives an x-independent $m_{3/2}$. - It leaves the plateau intact: a full supergravity check finds V changes by $\leq 0.6\%$ for $m_{3/2} \leq 10^{13}$ GeV. - It gives the angle a tachyonic mass $-2\, m_{3/2}^2$. Kähler-only stabilisers do not cure this: m_θ^2 stays $-2\, m_{3/2}^2$ for any $b(T-\bar{T})^4$ or $b(T-\bar{T})^2$ term (D270). A cure needs the uplift sector to couple to the angle, which makes the angle heavy (~$m_{3/2}$). - It needs $m_{3/2} \lesssim 6\times10^{8}$ GeV, so that the angle stays light while matter forms. - It needs $m_{3/2} \gtrsim 10^{8}$ GeV, so that gravitino decay releases $\lesssim 15\%$ of the energy. - It therefore works only for $m_{3/2} \approx 10^{8}$–10^{9} GeV (estimate), with R-parity violation.
>
> Neither option is derived. Adopt (i).

2. Section 2.2 (coupling): result of the alternative reading

Add to §2.2: “Coupling instead to $\partial(y/x)$ (the $1/\gamma_{BI} = y/x$ reading) lowers the matter yield by a factor 0.44 (D268/D297). The fitted hidden mass then moves from $m_\sigma \approx 7.5\times10^{10}$ to $\approx 3.0\times10^{10}$ GeV. The mechanism works under either reading.”

3. Table of assumptions, item 4

Replace the item with: “Supersymmetry breaking: a near-massless gravitino with superpartner masses not tied to m_3/2 (adopted); $W \propto T^{3/2}$ plus uplift needs the uplift sector to stabilise the angle and works at best for $m_{3/2} \approx 10^8$–10^9 GeV; a constant W_0 is excluded.”

4. Section 5.1 (fifth force): add precedents (D273)

Add after the shield paragraph:

> In generic supergravity quintessence, every MSSM soft term and fermion mass depends on the field through $e^{K/2}$, and Cassini then requires a tuned cancellation [Brax & Martin, hep-th/0605228]. The shield must therefore also keep the Standard Model’s supersymmetry-breaking scale independent of x. The no-scale isometry protects the field’s own lightness from gravitino-induced masses [Brax, Davis & Sakstein, arXiv:1212.4392], but not its couplings to matter.

5. Exploratory note §1.3–1.4: locked PC results (D277)

- **m_σ.** Replace “confirmation pending” with: “the locked confirmation run gives 1.25× observed at $m_\sigma = 5.8\times10^{10}$ GeV, the same in TWIST-K; entropy dilution ×4.6”.
- **§1.4 Isocurvature.** Replace “unverified” with: “the locked sensitivity run gives $|d \ln Y/d \ln \sigma_0| = 3.70$, which passes the locked limit of 5. $S_B \approx 3.7\times10^{-5}$ against a bound of 5×10^{-5} at $f = 0.1\ M_P$ is marginal, and requires $f \gtrsim 0.1\ M_P$. The yield is very sensitive to the angle’s starting value ($y_0 = 0.01$ multiplies it by about 9000), so the heavy angle during inflation is essential.” The earlier “8× below the bound” is withdrawn.

6. Gravitino consequences (D279)

Add to §5.2 option (i): “The gravitino is a thermal warm relic, making up $m_{3/2}/86$ eV ≈ 1–5% of the dark matter, with $\Delta N_{eff} \approx 0.02$–$0.05$. The main dark-matter candidate is unaffected.”

7. A new stated assumption: the reheating field χ has modular weight 3/2 (D286)

Add to the assumptions list:

> **The reheating field χ enters the Kähler potential as $|\chi|^2/(T + \bar{T})^{3/2}$.** Its physical mass is then independent of x, and its one-loop correction to the plateau is negligible, as D156 assumes.
>
> The other placements fail: - Inside the no-scale log, the one-loop term would be an order-one $\propto 1/x$ deformation ($dV/V \approx 7$ at the CMB point). - With a flat Kähler term, it would be larger still.

8. Option (i) needs a non-minimal SUSY-breaking sector (D291)

Add to §5.2:

> The 0.8–4.7 eV window is open only in non-minimal models, such as vector-mediated D-term breaking [Hook, McGehee & Murayama, arXiv:1801.10160] or strongly coupled messengers [Lu & Shirai, arXiv:1712.02359]. Minimal gauge mediation is excluded. A cosmic-shear bound below ~1 eV would remove this option.

9. A supersymmetry-breaking sector with x-independent scales (D292; supersedes the two-option text in item 1)

Replace the §5.2 options with:

> **$W \supset w\, T^{3/2} + \mu^2 X$, with the Polonyi field X at modular weight 3:** $K \supset |X|^2 (T + \bar{T})^{-3} [1 + c((T - \bar{T})/(T + \bar{T}))^2]$.
>
> - Along real T, exactly: $V = \mu^4 - 3w^2/8$, the same for every x. A single cosmological-constant tuning sets it to zero.
> - $m_{3/2} = w/2^{3/2}$ and $F_X = \mu^2$, both x-independent.
> - The angle mass is $(4c - 2)\, m_{3/2}^2$, so the angle is stable for $c > 1/2$.
> - With messengers at modular weight 3/2, the gauge-mediated soft masses do not slide. This removes the α_{em} drift of D289.
> - The eV gravitino window (D291) is realised in this sector.
> - The modular weights are assumed, not derived.

10. Condition on the SUSY-breaking sector (D294)

Add to §5.2:

> $X = 0$ is shifted by a Polonyi tadpole to $X \approx 2\sqrt{6}/(12q - 2)\, M_P$. That shift is x-independent in canonical units. The sector must also give the sgoldstino a mass far above $m_{3/2}$ ($q \gtrsim 10^7$); otherwise its late oscillations overclose the universe. This holds for dynamical SUSY breaking at $\sqrt{F} \sim 10^5$ GeV, but not for a bare Polonyi field.

11. Angle relic in the eV-gravitino branch (D298–D299)

Add to the exploratory note:

> With the late angle mass from §5.2 (~1–5 eV), the angle's leftover displacement after matter formation ($\varphi_i \approx 1.1 \times 10^{11}$ GeV, computed) gives a cold, ALP-like relic. It makes up roughly 12–28% of the dark matter (order-of-magnitude estimate), so it does not overclose. This is a consequence that could be tested; it is not yet locked.

12. Literature framing for the string-home limitation (D302, D304)

Add to §7:

> No controlled string regime is known for a thawing tail like this one. Cicoli et al. (Fortsch. Phys. 2022) find no slow-roll region in the parametrically controlled tree-level type II or heterotic moduli space; heterotic runaways give ε = 4. For minimal Z_3/Z_7 orbifolds, the modulus-independent gauge couplings needed by the shield are excluded by duality-anomaly matching [Ibáñez & Lüst, hep-th/9202046], so they require extra charged matter.

13. Why the tail field cannot be a heterotic volume modulus (D309)

Add to §7:

> In the weakly coupled heterotic string, α_GUT = e^{2φ} α′³/(16πV) [Witten, hep-th/9602070]. Keeping the gauge couplings fixed while the volume grows (as the shield requires) therefore drives the string coupling strong. The Twist's x cannot be a heterotic volume modulus. The strong-coupling continuation, heterotic M-theory, moves the Standard Model coupling along its matter-neutral direction (D208).

14. Exploratory note: a candidate string home (papers/twist_v5/G2_HOME_PROPOSAL.md, D309–D315)

Add a short section summarising the G_2 block-split proposal, and list question 12 among the open questions for string theorists.

15. n_s inconsistency is background-conditional (D223, D329)

Add to the n_s discussion:

> The NS-SPA inconsistency arises from CMB + BAO jointly. For Planck + DESI, freeing (w_0, w_a) returns n_s = 0.9638 ± 0.0038, the CMB-only value [arXiv:2512.05108]. The Twist's own, milder dark-energy evolution does not do this: our NS-SPA fit with the Twist background still prefers n_s ≈ 0.973–0.976. The verdict therefore depends on how the BAO–CMB tension resolves (DESI DR3, Simons Observatory).

16. The shield as a one-loop no-go (D323–D334)

Replace the shield paragraph's "assumption" wording with a precise statement:

> If the dark-energy field X enters as $K \supset -3 \ln(X + \bar{X})$ (which makes the plateau and the tail), the holomorphic f and Yukawas are X-independent, and the matter metrics are universal, then at one loop the Standard Model's Yukawas and gauge couplings cannot both be X-independent. Keeping Yukawas fixed gives $d(1/\alpha_a)/d \ln x = -b_a/4\pi$ [Kaplunovsky–Louis; local-model form in arXiv:0901.4350; general hidden-sector statement in arXiv:1811.10633], so $d \ln \alpha_{em}/d \ln x \approx 7\times10^{-3}$. Non-universal weights with a Giudice–Masiero μ need a Green–Schwarz term. The dilaton version of that shifts the no-scale α by ∓ 0.014 and spoils inflation (locked test). A twisted-moduli version passes inflation only with group-dependent couplings $\propto b_a$, which no known model provides. The shield is therefore an assumption beyond known string constructions, with a precisely stated requirement.

17. CMB-S4 cancelled (D346)

DOE and NSF ended support for CMB-S4 in July 2025. In the paper's tests table, replace "Simons Observatory, LiteBIRD, CMB-S4 class" with "Simons Observatory (and upgrades), BICEP Array, LiteBIRD". In the exploratory note §5, replace "LiteBIRD and CMB-S4" with "LiteBIRD and Simons Observatory". Add: with $r \approx 0.0036$, the expanded Simons Observatory (six SATs to 2035, $\sigma_r \approx 1.2\times10^{-3}$ baseline, arXiv:2512.15833) reaches ~3σ, and LiteBIRD ($\delta r = 0.001$, launch JFY 2032, arXiv:2406.02724) is comparable; an r decision is expected ~2032–35.

18. The split model TWIST-I (Danny's choice, 14 Nov 07:08; PREREG_TWISTI + D1–D2)

Present v6 with two readings. (a) **TWIST-I (adopted):** the Twist is the inflaton; its field rolls through matter genesis as in v5 and then settles at $x_{min} \geq 30$ and is stabilised with m_x (and the angle) $\gtrsim 100$ TeV so that it decays before BBN, and is captured adiabatically (residual displacement $\lesssim 10^{-7}$–10^{-5} M_P, D359–D360a), since a non-adiabatic arrival dilutes n_B/s and DM by 10^3–10^{13} (PREREG_TWISTI D1–D4); dark energy is Λ. Cost: the stabiliser minimum must coincide with the point where x comes to rest to within ~10^{-7}–10^{-5} M_P (D362), a third tuning besides Λ and n — unless x is trapped at an enhanced symmetry point (ESP) at $x \gtrsim 30$, crossed at $T \lesssim 10^{10}$–10^{12} GeV with coupling $g \gtrsim 0.1$, where the stabiliser has its minimum by symmetry (moduli trapping, hep-th/0403001; D363–D364). A χ-sector ESP alone is radiatively a maximum, so a separate stabiliser with its minimum at the ESP by symmetry is still needed (D366). Capturing late makes x's kinetic energy harmless. This removes the shield problem (item 16) and the H_0 cost, and gives up the tail predictions and "Λ from the cliff". (b) **The original tail** is kept as a conditional variant, valid only if the shield no-go (item 16) is evaded. The tests table splits accordingly: r, n_s and β apply to both; $w = -1$ applies to TWIST-I; the tail tests (WATCH-PURE-DR3, WA-CEILING, GROWTH, E_G) apply only to (b).

19. TWIST-I's CMB predictions are Starobinsky-class (D368)

Add to the TWIST-I description: “At N* = 55.7–57.6, the Twist’s n_s and r agree with the α = 1 attractor formulas ($n_s = 1 - 2/N$, $r = 12/N^2$) to within 10^{-4} and 3×10^{-4}. The inflationary test therefore probes the plateau class and, through N, *the Twist’s reheating history (its built-in kination phase raises N* to 55.7–57.6, toward but not into the $N^* \gtrsim 58$ that refined Starobinsky analyses need for ACT compatibility, arXiv:2504.20757; D370). The model-specific content is in N*, the matter mechanism and its isocurvature, and the dark-matter window.”